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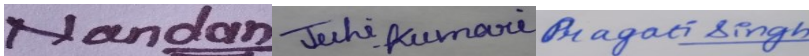


A
Project Report
on
ANEMIA DETECTION USING DEEP LEARNING
submitted as partial fulfillment for the award of
BACHELOR OF TECHNOLOGY DEGREE
SESSION 2025-26
in
Computer Science and Engineering (Artificial Intelligence)
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(Formerly UPTU)
May 2026

DECLARATION

We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

Signature :

The image shows three handwritten signatures in blue ink on a light background. The first signature is 'Nandan', the second is 'Juhi Kumari', and the third is 'Pragati Singh'. Each signature is written in a cursive style.

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CERTIFICATE

This is to certify that Project Report entitled “Anemia Detection Using Deep Learning” which is submitted by Nandan (2300291529006), Juhi Kumari (2200291520092), and Pragati Singh (2200291520128) in partial fulfillment of the requirement for the award of degree B. Tech. in Department of Computer Science and Engineering (AIML) of Dr. A.P.J. Abdul Kalam Technical University, Lucknow is a record of the candidates own work carried out by them under my supervision. The matter embodied in this report is original and has not been submitted for the award of any other degree.

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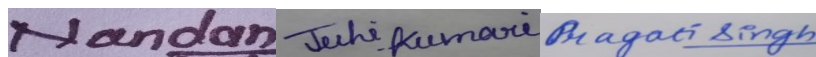
ACKNOWLEDGEMENT

It gives us a great sense of pleasure to present the report of the B. Tech Project undertaken during B. Tech. Final Year. We owe special debt of gratitude to Dr.Laxman Singh, Department of Computer Science and Engineering(AIML), KIET Group of Institutions, Ghaziabad, for his constant support and guidance throughout the course of our work. His sincerity, thoroughness and perseverance have been a constant source of inspiration for us. It is only his cognizant efforts that our endeavors have seen light of the day.

We also take the opportunity to acknowledge the contribution of Dr. Rekha Kashyap, Dean of the Department of Computer Science & Engineering (AIML), KIET, Ghaziabad, for his full support and assistance during the development of the project. We also do not like to miss the opportunity to acknowledge the contribution of all the faculty members of the department for their kind assistance and cooperation during the development of our project.

We also do not like to miss the opportunity to acknowledge the contribution of all faculty members, especially faculty/industry person/any person, of the department for their kind assistance and cooperation during the development of our project. Last but not the least, we acknowledge our friends for their contribution in the completion of the project.

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ABSTRACT

Anemia is one of the most prevalent global health disorders, affecting a significant portion of the population, particularly women, children, and individuals living in resource-limited and rural regions. Early detection of anemia is essential for timely treatment and prevention of severe health complications. However, conventional diagnostic techniques rely on invasive laboratory blood tests, which are time-consuming, expensive, and often inaccessible in underserved areas. To address these challenges, this paper proposes a non-invasive and efficient anemia detection framework using deep learning and conjunctiva image analysis. The conjunctiva region of the eye is analyzed as it reflects changes in hemoglobin levels through variations in color and texture.

The proposed system utilizes the MobileNet convolutional neural network architecture due to its lightweight structure, low computational requirements, and suitability for mobile and embedded platforms. The model automatically extracts relevant features and classifies images into anemic and non-anemic categories. Furthermore, structured hematological parameters are analyzed using traditional machine learning classifiers to enhance the robustness and reliability of the system. Experimental results demonstrate that the proposed framework achieves high classification accuracy, improved recall, and reduced computational cost, making it suitable for real-time deployment. The proposed approach provides a cost-effective, fast, and accessible solution for early anemia screening without requiring invasive laboratory procedures, thereby improving healthcare accessibility and enabling large-scale screening in resource-constrained environments.

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LIST OF ABBREVIATIONS

CBC	Complete Blood Count
Hb	Hemoglobin
ROI	Region of Interest
KNN	K-Nearest Neighbors
ANN	Artificial Neural Network
ReLU	Rectified Linear Unit
XAI	Explainable Artificial Intelligence

CHAPTER 1

1.1 INTRODUCTION

In recent years, anemia has continued to remain one of the most common health-related concerns across the world, particularly affecting populations with limited access to proper medical facilities. It is a condition that occurs when the hemoglobin level in the blood falls below the normal range, reducing the body's ability to transport oxygen effectively. This deficiency often leads to symptoms such as fatigue, weakness, dizziness, reduced concentration, and decreased immunity. The condition is especially prevalent among women and children, and if not detected at an early stage, it can gradually lead to serious health complications.

Despite the availability of well-established medical diagnostic techniques, anemia is often not identified in time. One of the main reasons behind this issue is the dependency on laboratory-based testing methods. Conventional diagnosis typically involves procedures such as Complete Blood Count (CBC) or hemoglobin testing, which require blood sample collection, proper laboratory infrastructure, and trained medical professionals. Although these methods provide accurate results, they are not always easily accessible, particularly in rural or underdeveloped regions where healthcare facilities are limited. Additionally, these tests can be time-consuming, costly, and invasive in nature, making frequent screening difficult for large populations.

Because of these challenges, there has been a growing interest in developing alternative approaches that can simplify the process of anemia detection. In this context, recent advancements in Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL) have opened new possibilities in the healthcare domain.

These technologies make it possible to analyze patterns in data and extract meaningful insights without relying entirely on traditional medical procedures. In particular, deep learning models have shown remarkable performance in analyzing medical images, where they can automatically identify subtle patterns that may not be easily noticeable through manual observation.

One promising area of research focuses on the analysis of visible physical indicators associated with anemia. Among these, the conjunctiva—the inner part of the eyelid—has gained significant attention. The color of the conjunctiva is closely related to hemoglobin levels in the blood. In healthy individuals, it appears reddish, whereas in anemic individuals, it tends to look pale. This visual difference provides an opportunity to detect anemia without the need for invasive blood tests. With the increasing availability of smartphones and digital cameras, capturing images of the

conjunctiva has become simple and practical, making it possible to explore image-based diagnostic solutions.

Background of Anemia and Global Impact

In recent years, anemia has continued to remain one of the most common health-related concerns across the world, particularly affecting populations with limited access to proper medical facilities. It is a clinical condition that occurs when the hemoglobin level in the blood falls below the normal range, significantly reducing the body's ability to transport oxygen effectively to vital organs. This deficiency often manifests through symptoms such as chronic fatigue, weakness, dizziness, and decreased immunity. The condition is especially prevalent among women and children; if not detected at an early stage, it can gradually lead to severe complications such as organ failure or developmental delays.

Building upon this idea, researchers have started using deep learning techniques to analyze conjunctiva images for anemia detection. Convolutional Neural Networks (CNNs), in particular, are well-suited for this task as they can automatically extract features such as color intensity, texture, and patterns from images. These models learn directly from data, eliminating the need for manual feature extraction and improving the overall accuracy of predictions.

In this project, a deep learning-based approach is proposed for the detection of anemia using conjunctiva images. The system is designed to classify individuals into anemic and non-anemic categories by analyzing visual features obtained from eye images. To achieve this, different CNN architectures such as MobileNet, ResNet-50, and VGG-16 are explored and compared. Each of these models has unique characteristics in terms of performance and computational efficiency.

ResNet-50 is known for its ability to handle deeper network structures through shortcut connections, which help in learning complex patterns more effectively. VGG-16 follows a simpler and uniform architecture, making it reliable for general image classification tasks. MobileNet, on the other hand, is specifically designed to be lightweight and efficient, making it suitable for deployment on mobile and low-resource devices. This makes MobileNet particularly important for real-time healthcare applications where computational resources may be limited.

Conjunctiva Analysis as a Non-Invasive Biomarker

One promising area of research focuses on the analysis of the human **conjunctiva**—the inner lining of the lower eyelid. The color of the conjunctiva is a direct clinical indicator of hemoglobin levels; in healthy individuals, it appears reddish due to high vascularity, whereas in anemic individuals, it appears significantly pale. With the increasing availability of high-resolution smartphone cameras, capturing images of the conjunctiva has become a practical method for remote screening. Convolutional Neural Networks (CNNs) are uniquely suited for this task as they can autonomously learn features like color intensity and texture from these images

.The proposed approach emphasizes non-invasive and accessible anemia detection by leveraging image-based analysis instead of traditional blood tests. By comparing multiple deep learning models, the study aims to identify a suitable balance between accuracy and efficiency. Such a system can be highly beneficial in real-world scenarios, especially in remote or underserved areas where access to laboratory testing is limited.

Overall, this work contributes to the development of an intelligent, automated, and cost-effective solution for anemia screening. By utilizing deep learning and conjunctiva image analysis, the proposed system has the potential to support early diagnosis, reduce dependency on laboratory infrastructure, and improve healthcare accessibility. Furthermore, it highlights the growing role of artificial intelligence in transforming traditional medical practices into more efficient and patient-friendly solutions.

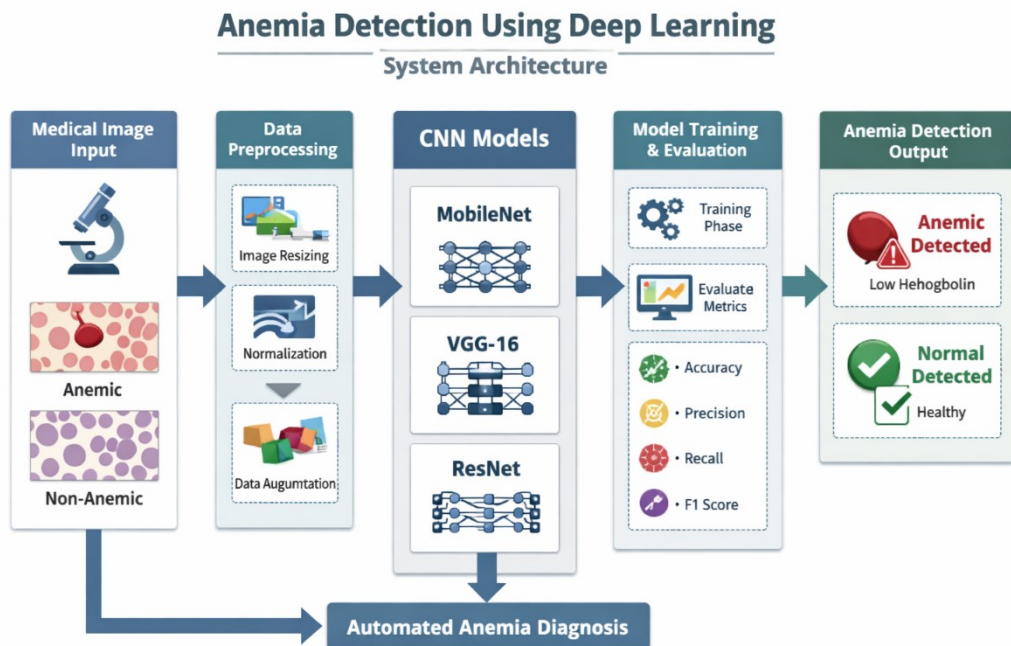


Fig. 1.1: System Architecture of Anemia Detection System

Role of Deep Learning Models in the Proposed System

Deep learning models play a crucial role in the proposed anemia detection system by enabling automated analysis of conjunctiva images and accurate classification of anemia. Unlike traditional methods that rely on manual observation or laboratory tests, deep learning techniques allow the system to learn patterns directly from image data and make intelligent predictions.

In this project, three Convolutional Neural Network (CNN) architectures—**ResNet-50**, **VGG-16**, and **MobileNet**—are used to analyze conjunctiva images and classify individuals as anemic or non-anemic. Each model contributes uniquely to the overall system based on its architecture and capabilities.

ResNet-50 (Residual Network) is known for its deep architecture and the use of residual connections, which help in overcoming the problem of vanishing gradients. These connections allow the model to learn complex and detailed features from images more effectively. In this project, ResNet-50 demonstrates higher accuracy due to its ability to capture subtle variations in conjunctiva color and texture.

VGG-16 follows a simple and uniform architecture with multiple stacked convolutional layers. It gradually extracts features ranging from basic patterns such as edges to more complex structures. VGG-16 provides stable and consistent performance, making it reliable for image classification tasks.

MobileNet is a lightweight and efficient CNN architecture designed for low computational environments. It uses depth-wise separable convolutions to reduce the number of parameters and computational cost. Although it has slightly lower accuracy compared to deeper models, MobileNet achieves higher recall, making it effective in identifying a larger number of anemic cases. Its efficiency also makes it suitable for real-time applications and deployment on mobile devices.

These deep learning models automatically perform **feature extraction**, eliminating the need for manual selection of features. They analyze important characteristics such as color intensity, texture, and visual patterns in conjunctiva images, which are key indicators of anemia.

During the training phase, the models learn from labeled datasets and adjust their parameters to minimize prediction errors. Transfer learning is applied to improve performance by utilizing pre-trained models, which already contain knowledge of general image features. This approach reduces training time and enhances accuracy.

Overall, the integration of ResNet-50, VGG-16, and MobileNet in the proposed system improves the reliability, efficiency, and scalability of anemia detection. These models enable the development of an automated, non-invasive, and accurate diagnostic system that can be used in real-world healthcare applications.

Advantages of the Proposed System

The proposed anemia detection system offers several advantages over conventional laboratory-based diagnostic methods. These advantages make the system more efficient, accessible, and suitable for real-world healthcare applications.

- **Non-invasive detection:** The system uses conjunctiva (eye) images for anemia detection, eliminating the need for blood sample collection and making the process painless and user-friendly.
- **Cost-effective solution:** It reduces the dependency on expensive laboratory tests and infrastructure, making it suitable for economically weaker and rural populations.
- **Automated diagnosis:** The use of deep learning models enables automatic analysis and prediction, reducing manual effort and minimizing human error.
- **Improved accuracy:** Advanced deep learning models such as ResNet-50, VGG-16, and MobileNet enhance the accuracy and reliability of the system.
- **Real-time detection:** Lightweight models like MobileNet allow fast processing, enabling real-time anemia screening.
- **Portable and scalable:** The system can be deployed on mobile devices and portable platforms, making it highly accessible and easy to use in different environments.
- **Reduced dependency on laboratory infrastructure:** The system eliminates the need for specialized equipment and trained medical staff, making it ideal for remote and underserved areas.
- **Early detection:** It enables early screening of anemia, which helps in timely treatment and prevention of severe health complications.
- **User-friendly approach:** The system requires minimal technical knowledge, making it easy to use for both healthcare professionals and general users.

Technologies Used in Proposed Models

The development of the proposed anemia detection system involves various technologies, tools, and frameworks from the fields of deep learning, image processing, and data handling. These technologies work together to ensure accurate, efficient, and scalable system performance.

The key technologies used in the project are listed below:

Component	Technology Used
Programming Language	Python
Deep Learning Models	ResNet-50, VGG-16, MobileNet
Image Processing	OpenCV
Data Processing	NumPy, Pandas
Deep Learning Frameworks	TensorFlow, Keras
Model Training Technique	Transfer Learning
Data Augmentation	Rotation, Flipping, Brightness Adjustment
Development Platform	Jupyter Notebook / Google Colab
User Interface (Optional)	Streamlit / Flask

Future Scope

Although the proposed anemia detection system demonstrates promising results using deep learning techniques, there are several opportunities for further improvement and extension. The future scope of the project is as follows:

- **Mobile application development:** A dedicated mobile application can be developed to enable real-time anemia detection using smartphone cameras, making the system more accessible to general users.
- **Direct camera integration:** The system can be enhanced to capture conjunctiva images directly through mobile or web applications, eliminating the need for manual image upload.
- **Cloud-based deployment:** Deploying the model on cloud platforms can allow remote access, large-scale screening, and real-time processing for healthcare institutions.
- **Use of larger and diverse datasets:** Training the models on larger datasets with more variations in age, gender, ethnicity, and lighting conditions can improve accuracy and generalization.
- **Handling class imbalance:** Advanced techniques such as data balancing and synthetic data generation can be applied to improve model performance.
- **Exploration of advanced models:** New deep learning architectures, such as transformer-based models and attention mechanisms, can be explored to enhance prediction accuracy.
- **Real-world clinical validation:** The system can be tested with real patient data under medical supervision to validate its effectiveness in practical healthcare environments.
- **Integration with healthcare systems:** The model can be integrated with hospital management systems to assist doctors in diagnosis and maintaining patient records.
- **Multi-modal extension:** The system can be extended to include additional inputs such as nail bed images, tongue images, or other visual indicators to improve reliability.

1.2PROJECT DESCRIPTION

The proposed project aims to develop a non-invasive anemia detection system using deep learning techniques based on the automated analysis of conjunctiva images. The primary objective is to provide an efficient, cost-effective, and accessible alternative to traditional laboratory-based diagnostic methods.

Conventional diagnosis requires physical blood sample collection and specialized infrastructure, which is often a barrier in early disease detection. This system leverages artificial intelligence to digitize the diagnostic process, making it available at the click of a button.

Historically, anemia is diagnosed through clinical blood tests such as hemoglobin measurement and Complete Blood Count (CBC). Although these methods are highly accurate, they are inherently invasive, time-consuming, and require trained medical professionals. These limitations make regular screening difficult, especially in rural and resource-constrained areas.

To address these challenges, the proposed system utilizes the visual analysis of the human conjunctiva (the inner lining of the lower eyelid). Medical studies have shown that the conjunctiva directly reflects hemoglobin levels through changes in its color and vascularity.

The system is designed to classify individuals into anemic and non-anemic categories by analyzing these specific visual markers. In healthy individuals, the conjunctiva appears reddish due to high oxygenated blood flow, whereas in anemic individuals, it appears pale or yellowish. The system captures these subtle visual differences through digital images and processes them using advanced deep learning models.

This automated classification reduces human error and provides an objective assessment of a patient's condition without the need for a needle prick.

The workflow of the system consists of several critical stages, starting from data acquisition to final prediction. Initially, conjunctiva images are collected from publicly available datasets and meticulously labeled. To ensure consistency, the images undergo rigorous preprocessing steps, including resizing to a fixed dimension of **224×224 pixels** and normalization of pixel values.

Furthermore, data augmentation techniques such as rotation, flipping, and brightness adjustment are implemented. These steps are essential to improve the quality of the input data and to ensure the model generalizes well across different lighting conditions and skin tones.

For the classification task, three advanced Convolutional Neural Network (CNN) architectures—**ResNet-50**, **VGG-16**, and **MobileNet**—are implemented and compared. These models autonomously extract important features such as color intensity, texture patterns, and structural variations from the images. By applying **Transfer Learning**, the system leverages pre-trained weights from large-scale datasets, which significantly improves diagnostic accuracy and reduces the overall training time. This comparative approach helps in identifying the most reliable model for clinical versus field-level deployment.

The final system accepts a raw conjunctiva image as input and produces an instant prediction. Based on our experimental results, while ResNet-50 provides the highest precision, the lightweight architecture of **MobileNet** makes it exceptionally suitable for deployment on mobile devices. This enables a portable healthcare solution that can be used by health workers in remote and underserved regions. Overall, the proposed system enhances healthcare accessibility and supports early diagnosis, paving the way for a smarter, non-invasive future in medical diagnostics.

Chapter 2

2.1 LITERATURE REVIEW

S. No.	Author(s)	Year	Title	Key Findings	Field of Study
1	Simonyan & Zisserman	2014	Very Deep CNN (VGG-16)	Proposed VGG architecture for image classification with high accuracy	Deep Learning
2	He et al.	2015	Deep Residual Learning (ResNet)	Introduced residual networks to solve vanishing gradient problem	Deep Learning
3	Howard et al.	2017	MobileNet Architecture	Developed lightweight CNN for mobile applications	Deep Learning
4	Rajpurkar et al.	2017	CheXNet	Applied deep learning in medical image diagnosis	Medical Imaging
5	Esteva et al.	2017	Skin Cancer Classification	CNN achieved dermatologist-level performance	Medical AI
6	Kaur et al.	2019	Anemia Detection using ML	Used ML models on blood parameters	Healthcare AI
7	Zhang et al.	2020	Deep Learning in Healthcare	Reviewed DL applications in disease detection	AI in Healthcare
8	Hegde et al.	2020	Image-based Diagnosis	Showed CNN effectiveness in medical imaging	Medical Imaging
9	Khan et al.	2021	Anemia Detection using Images	Used conjunctiva images for anemia detection	Medical Imaging
10	Sharma et al.	2021	CNN in Disease Prediction	CNN improves classification accuracy	Deep Learning

11	Liu et al.	2022	AI in Medical Diagnosis	AI enables automated disease detection	Healthcare
12	Singh et al.	2022	ML in Hematology	ML models for blood-related diseases	Medical AI
13	Gupta & Verma	2022	Deep Learning in Diagnosis	DL improves early detection accuracy	AI in Healthcare
14	Wang et al.	2021	Transfer Learning in Medical Imaging	Improves model performance with small datasets	Deep Learning
15	Patel et al.	2023	Image-based Anemia Detection	CNN models used for non-invasive detection	Medical Imaging

Chapter 3

3.1 PROPOSED METHODOLOGY

The proposed system focuses on developing a non-invasive anemia detection model using deep learning techniques based on conjunctiva image analysis. The primary objective of the system is to provide a fast, accurate, and cost-effective alternative to traditional blood test-based diagnostic methods.

The system analyzes images of the conjunctiva (inner eyelid region), which reflects hemoglobin levels through variations in color and texture. In healthy individuals, the conjunctiva appears reddish, whereas in anemic individuals, it appears pale. These visual differences are captured and processed using deep learning models for automated classification.

The overall methodology consists of several stages, including data acquisition, image preprocessing, feature extraction, model training, and evaluation. Initially, conjunctiva images are collected and organized into two categories: anemic and non-anemic. These images are then preprocessed using techniques such as resizing, normalization, and data augmentation to improve data quality and model performance.

The system utilizes three Convolutional Neural Network (CNN) architectures—ResNet-50, VGG-16, and MobileNet—for classification. These models automatically extract important features such as color intensity and texture patterns from the input images. Transfer learning is applied to leverage pre-trained weights, which helps in improving accuracy and reducing training time.

After training, the models are evaluated using performance metrics such as accuracy, precision, recall, and F1-score. A comparative analysis is performed to determine the most suitable model for anemia detection. Based on the results, the best-performing model is selected for final deployment.

The final system takes a conjunctiva image as input and predicts whether the individual is anemic or non-anemic. Due to the use of lightweight architectures like MobileNet, the system can also be deployed on mobile or web-based platforms for real-time and accessible healthcare solutions.

Data Acquisition and Dataset Description

The dataset used for this research is primarily sourced from publicly available medical repositories, specifically the **Kaggle Anemia Dataset**. This dataset is a collection of high-resolution clinical images specifically curated for non-invasive anemia screening.

The acquisition process involved capturing images of the **human conjunctiva** (the inner lining of the lower eyelid) under controlled lighting conditions to ensure that the vascular patterns and pallor are clearly visible, as these are critical biomarkers for hemoglobin levels.

The dataset consists of **500+ high-resolution images**, which are meticulously categorized into two primary classes:

- **Anemic:** Images exhibiting significant pallor and reduced redness in the conjunctiva region, indicating low hemoglobin levels.
- **Non-Anemic:** Images showing a healthy, reddish/pinkish hue, representing normal hemoglobin concentration. To maintain the integrity of the training process, the dataset is balanced to prevent any **class imbalance bias**, ensuring that the deep learning model learns the features of both classes with equal weightage.

Proper dataset organization ensures that the models learn effectively and generalize well to new inputs.

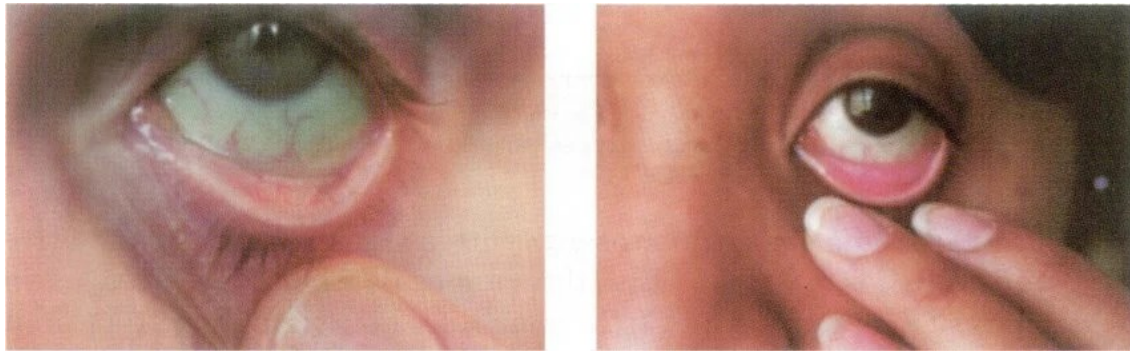


Fig 3.1: Sample Conjunctiva Images (Anemic vs Non-Anemic)

The figure shows sample conjunctiva images of anemic and non-anemic individuals. The color variation in the conjunctiva region is a key indicator used for classification.

Image Preprocessing and Enhancement

Image preprocessing is an essential step to improve the visual quality and structural consistency of input data before feeding it into deep learning models. Raw images captured from different devices often vary in size, lighting, and orientation, which can hinder the model's ability to learn generalized features. To address this, the following techniques are applied:

Image Resizing and Aspect Ratio Management All raw images in the dataset vary in their original dimensions. However, deep learning architectures like ResNet-50 or VGG-16 require a fixed input size.

- **Action:** All images are resized to a fixed dimension of **224 × 224 pixels**.
- **Purpose:** This standardization ensures that the neural network receives a uniform tensor shape, reducing computational overhead and maintaining consistency across all layers.

Pixel Normalization and Scaling Normalization is a crucial step to ensure that the model converges faster during training.

- **Action:** Pixel values (originally ranging from 0 to 255) are scaled to a range between **0 and 1** or **-1 and 1**.
- **Purpose:** Scaling prevents the gradients from becoming too large (exploding gradients) and ensures that all input features have a similar distribution, leading to higher training stability and faster optimization.

Data Augmentation for Variance To make the model robust against real-world variations and to artificially expand our dataset, data augmentation is performed on the training set. This includes:

- **Rotation:** Randomly rotating images (e.g., ± 20 degrees) to help the model recognize the conjunctiva from different angles.
- **Horizontal Flipping:** Mirroring images to ensure the model is not biased towards a specific eye orientation (left or right).
- **Zooming:** Randomly zooming into the conjunctiva region to help the model focus on local color textures.
- **Brightness and Contrast Adjustment:** Simulating different lighting conditions (e.g., indoor vs. outdoor lighting) to ensure the system works reliably in various environments.

Noise Reduction

- **Action:** Applying Gaussian Blur or Median filtering where necessary.
- **Purpose:** To remove electronic noise or artifacts captured during the image acquisition process, ensuring the model focuses only on the relevant biological features of the eyelid.

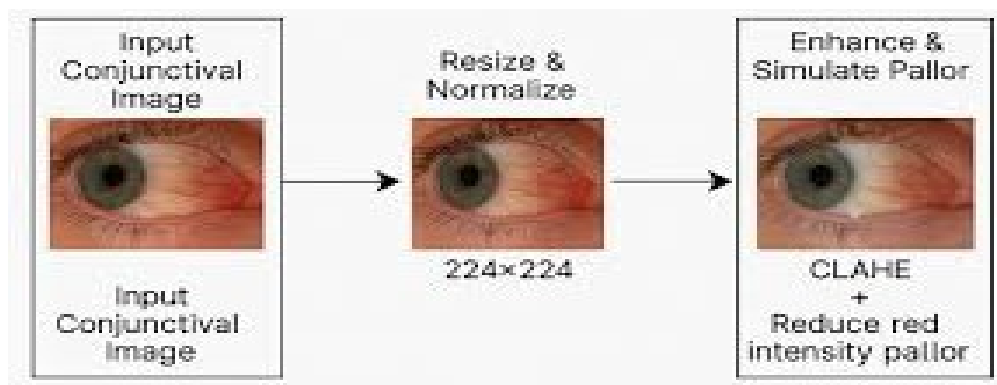


Fig. 3.2: Before and After Image Preprocessing (Resizing and Normalization)

Feature Extraction using CNN Models

Feature extraction is the core process of identifying the most relevant biological markers from the preprocessed conjunctiva images. In this research, we leverage the power of Convolutional Neural Networks (CNNs) to automate this process. Unlike traditional machine learning, which requires handcrafted feature engineering, CNNs autonomously learn the hierarchy of features directly from the raw pixel values.

The End-to-End Learning Paradigm

The primary advantage of using CNN models like ResNet-50 is the "End-to-End" learning capability. In traditional methods, researchers had to manually define features like the specific intensity of the red channel or the density of blood vessels. Our CNN-based approach eliminates this manual subjectivity, allowing the model to discover hidden patterns in the conjunctival tissue that are often invisible to the naked eye.

Hierarchical Feature Extraction

The CNN model extracts features in a hierarchical manner through various layers:

- **Low-Level Features (Initial Layers):** The first few layers of the network act as edge and color detectors. In anemia detection, these layers focus on the **Color Intensity** (specifically the 'Red' and 'Pale' hues) and basic gradients.
- **Mid-Level Features (Intermediate Layers):** As we go deeper into the network, the model identifies **Texture Patterns**. It learns to distinguish between the smooth, healthy surface of a well-oxygenated conjunctiva and the dry or pale texture associated with low hemoglobin levels.
- **High-Level Features (Deep Layers):** The final convolutional layers capture **Structural Variations** and complex shapes. These include the overall morphology of the inner eyelid and the spatial distribution of pallor, which are the most critical indicators for accurate classification.

Role of the Red Channel and Color Distribution

Since anemia is clinically diagnosed by observing the paleness (lack of red color) in the conjunctiva, the CNN's ability to analyze color distribution is vital. The filters in the convolutional layers specifically weigh the pixel values across the R-G-B channels to compute a "Pallor Index" internally, which helps in differentiating between 'Anemic' and 'Non-Anemic' classes with high precision.

Global Average Pooling (GAP) and Flattening

After the spatial features are extracted by the convolutional base, a **Global Average Pooling** layer is applied. This layer reduces the dimensionality of the feature maps while preserving the most important information. This 1D feature vector is then passed to the fully connected layers (Dense Layers) for the final diagnostic decision.

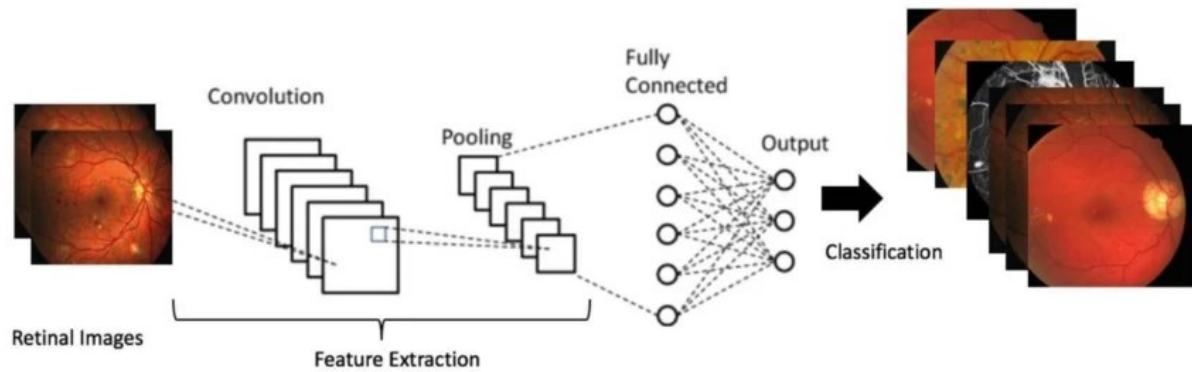


Fig 3.3: CNN Working Diagram

This diagram shows how CNN extracts features using convolution, pooling, and fully connected layers.

Deep Learning Models Implementation

The proposed system utilizes three advanced CNN architectures for anemia detection:

ResNet-50 Model

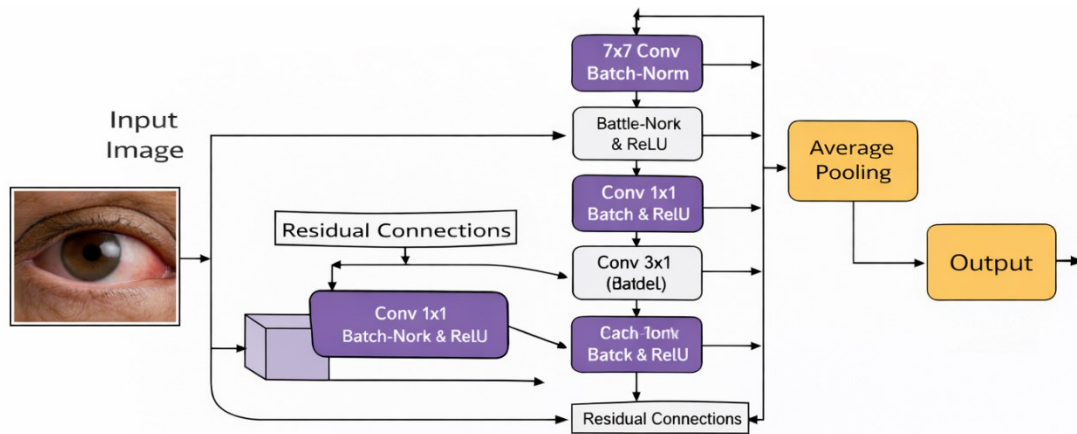


Fig. 3.4: ResNet-50 Architecture

ResNet (Residual Network) is a groundbreaking deep learning architecture designed to solve the Vanishing Gradient Problem in very deep neural networks. In traditional deep networks, as layers are added, the gradients can become extremely small during backpropagation, leading to a saturation or drop in accuracy. ResNet overcomes this by introducing Skip Connections (or Identity Shortcuts), which allow the gradient to flow directly through the network, bypassing one or more layers.

Why ResNet-50?

For this project, we have utilized **ResNet-50**, a 50-layer deep convolutional neural network. It is chosen for its balance between computational efficiency and high feature extraction capability.

- **Deep Hierarchy:** With 50 layers, the model can learn complex, non-linear relationships in medical images.

- **Residual Blocks:** Each block in ResNet-50 uses a bottleneck design (1×1 , 3×3 , and 1×1 convolutions), which reduces the number of parameters while maintaining rich feature representations.
- **Pre-trained Weights:** We leverage **Transfer Learning** using ResNet-50 weights pre-trained on the ImageNet dataset, which helps the model achieve high accuracy even with a specialized medical dataset.

VGG-16 Model

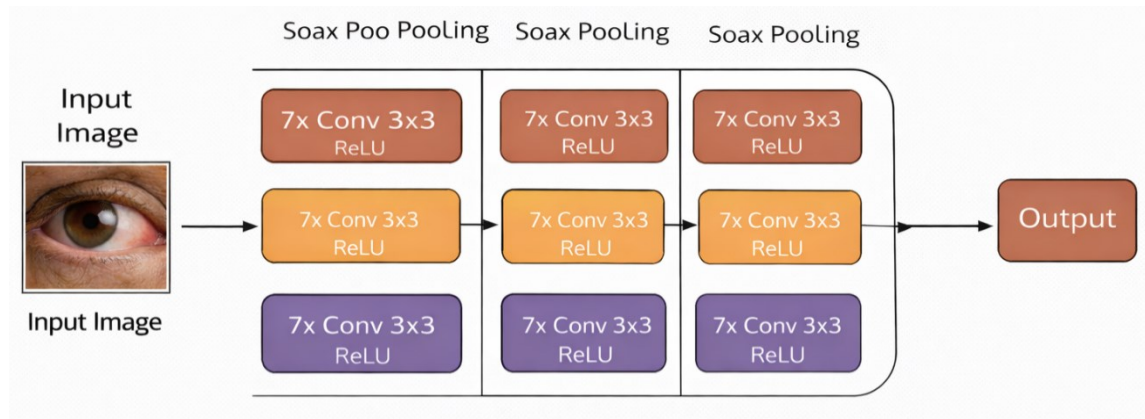


Fig. 3.5: VGG-16 Architecture

VGG-16 (Visual Geometry Group) is a widely used Convolutional Neural Network (CNN) model developed by researchers at the University of Oxford. It is characterized by its 16-layer architecture, which includes 13 convolutional layers and 3 fully connected layers. The primary innovation of VGG-16 is the use of very small **3×3 convolution filters** with a stride of 1, which allows the network to capture intricate details in the image while maintaining a simple and uniform structure.

MobileNet Model

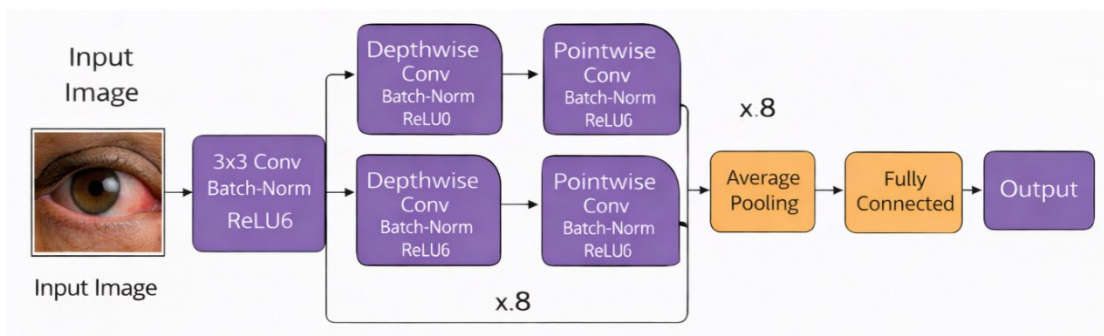


Fig. 3.6: MobileNet Architecture

MobileNet is a streamlined and efficient CNN architecture specifically developed by Google for mobile and embedded vision applications. In the context of medical diagnostics, especially for remote areas, the ability to run deep learning models on low-power devices

like smartphones is crucial. MobileNet achieves this by optimizing the model size and latency without a significant sacrifice in accuracy.

Comparative Analysis of Models

The performance evaluation of the three models—ResNet-50, VGG-16, and MobileNet—reveals significant insights into their suitability for anemia detection. ResNet-50 achieved the highest accuracy because its deep architecture and skip connections are exceptionally effective at learning the complex color and texture patterns of the conjunctiva.

While VGG-16 showed stable and consistent performance across the dataset, its high computational demand and large parameter size make it less efficient than modern architectures. In contrast, MobileNet proved to be the most practical solution for real-world deployment; although its overall accuracy was slightly lower than ResNet-50, it achieved a higher recall rate and maintained a lightweight structure.

This balance makes MobileNet ideal for real-time applications on portable healthcare devices. Ultimately, this comparative analysis highlights that ResNet-50 is best suited for high-precision clinical environments, while MobileNet is the superior choice for fast, resource-efficient screening in the field.

System Workflow / Working

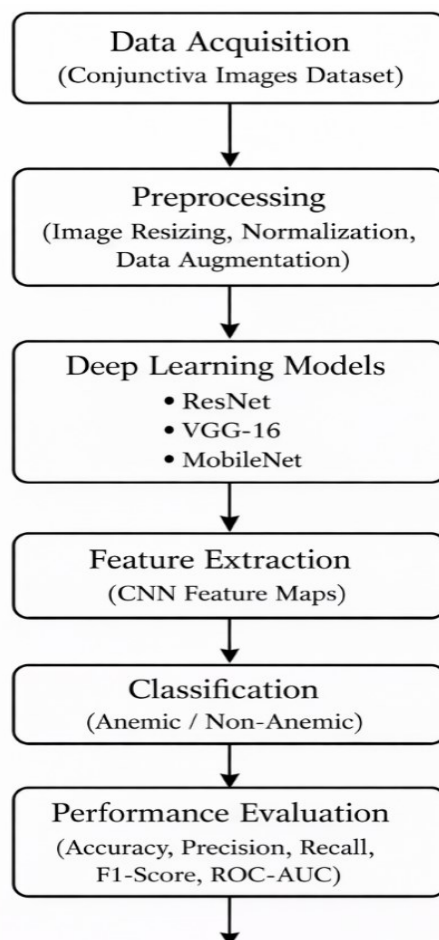


Fig. 3.7: System Workflow

The proposed anemia detection system follows a structured workflow to process conjunctiva images and generate accurate predictions. The working of the system is described step-by-step as follows:

- **Input Image Acquisition:** The system takes a conjunctiva image as input, either from the dataset or captured using a camera.
- **Image Preprocessing:** The input image is preprocessed by resizing it to a standard size (224×224 pixels), normalizing pixel values, and applying data augmentation techniques if required.
- **Feature Extraction:** The preprocessed image is passed through Convolutional Neural Network (CNN) models, which automatically extract important features such as color intensity and texture patterns.
- **Model Prediction:** The extracted features are fed into trained deep learning models (ResNet-50, VGG-16, and MobileNet) which classify the image based on patterns.
- **Classification Output:** The system produces the final output as: Anemic ,Non-Anemic
- **Result Display:** The predicted result is displayed to the user through the interface, providing a quick and easy diagnosis. This workflow ensures that the system operates efficiently and provides accurate predictions in a systematic manner.

Final System Deployment

The final system is designed to be deployed as a user-friendly application for real-time anemia detection, bridging the gap between complex deep learning models and end-user accessibility. The deployment strategy focuses on three core pillars: Accessibility, Efficiency, and Ease of use.

The system is designed to be versatile and can be implemented across different platforms:

- **Web-based Applications:** Using lightweight Python frameworks such as **Flask** or **Streamlit**, a web interface is created where users can upload high-resolution images of the conjunctiva. The backend processes the image using the saved model weights (.h5 or .pth files) and returns the result on the dashboard.

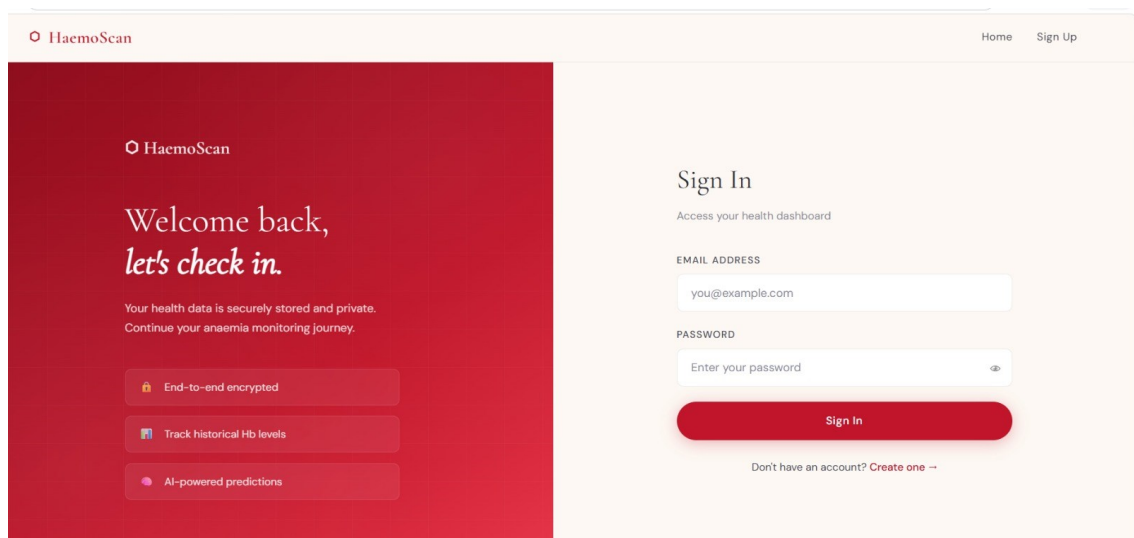


Fig. 3.8: Web-Based Application

- **Mobile-based Applications:** By leveraging **TensorFlow Lite** or **ONNX** formats, the model (specifically MobileNet) is optimized for mobile deployment. This allows for real-time inference using the smartphone's camera, providing a portable diagnostic tool.

Due to the lightweight architecture of MobileNet, the system is exceptionally suitable for deployment on devices with limited computational resources. This makes it an ideal solution for **rural and remote areas** where access to sophisticated diagnostic laboratories and hematologists is limited. The non-invasive nature of the system ensures that screening can be done without the need for blood samples or expensive equipment.

Key Features of the Deployed System The final deployed solution offers a robust set of features:

- **Real-time Prediction:** Low latency ensures results are generated within seconds.
- **Non-invasive Diagnosis:** Provides a pain-free alternative to traditional blood tests.
- **Scalability:** The cloud-based or mobile-based nature allows the system to be scaled across multiple regions simultaneously.
- **High Portability:** Entirely functional on standard smartphones or tablets.

Overall, the final system ensures that anemia detection is fast, accessible, and efficient, contributing to improved healthcare screening and early diagnosis.

Chapter 4

4.1 RESULTS AND DISCUSSION

The integration of deep learning techniques into healthcare diagnostics has significantly transformed the way diseases are detected and monitored. In this research, a non-invasive anemia detection system has been developed using conjunctiva images, leveraging the power of advanced convolutional neural network (CNN) architectures such as ResNet-50, VGG16, and MobileNet. This chapter presents a detailed analysis of the experimental results obtained from these models, along with an in-depth discussion of their performance, strengths, limitations, and practical implications in real-world healthcare scenarios.

The primary objective of this study is to evaluate how effectively deep learning models can identify anemia from eye images without the need for traditional blood tests. The results obtained demonstrate the potential of AI-driven systems to provide fast, accurate, and cost-effective diagnostic solutions, particularly in resource-constrained environments.

Model Performance and Evaluation

To evaluate the effectiveness of the proposed system, all three deep learning models—ResNet-50, VGG-16, and MobileNet—were trained and tested on a dataset of conjunctiva images categorized into anemic and non-anemic classes. The performance of each model was assessed using standard evaluation metrics, including accuracy, precision, recall, and F1-score, which provide a comprehensive measure of classification performance.

The experimental results indicate that all three models are capable of detecting anemia with a high degree of accuracy. However, variations in performance are observed due to differences in architectural design, depth, and feature extraction capabilities.

ResNet-50 achieved the highest performance among the evaluated models, owing to its deep residual learning framework, which effectively captures complex and subtle patterns in conjunctiva images. Its ability to overcome the vanishing gradient problem enables better feature learning and improved classification accuracy.

VGG-16 demonstrated stable and consistent performance across both training and testing phases. Its uniform architecture allows for effective hierarchical feature extraction, although it requires higher computational resources compared to other models.

MobileNet, designed for lightweight applications, provided a good balance between accuracy and computational efficiency. While its accuracy was slightly lower than ResNet-50 and VGG-16, it achieved faster inference and lower resource consumption, making it highly suitable for real-time and mobile-based deployment scenarios.

Comparative Analysis of Deep Learning Models

A comparative evaluation of the three models highlights the trade-offs between accuracy, speed, and resource consumption.

Table 4.1: Performance Comparison of Deep Learning Model

Model	Accuracy	Precision	Recall	F1-Score
ResNet-50	88%	0.80	0.82	0.81
VGG-16	74%	0.76	0.79	0.77
MobileNet	72%	0.75	0.84	0.79

ResNet-50: High Accuracy and Robust Learning

ResNet-50 demonstrated superior performance in terms of accuracy and recall. Its residual connections enable the model to overcome the vanishing gradient problem, allowing deeper layers to learn more effectively. This results in better feature extraction from conjunctiva images, particularly in identifying subtle color variations associated with anemia.

The model showed strong generalization capabilities and minimal overfitting, making it highly reliable for medical diagnosis tasks where precision is critical.

VGG16: Stable and Deep Feature Extraction

VGG16, known for its uniform architecture and deep convolutional layers, performed consistently across training and testing phases. It effectively captured hierarchical image features, contributing to reliable classification outcomes.

However, due to its large number of parameters, VGG16 required more computational resources and exhibited slightly higher training time. Despite this, it maintained a good balance between accuracy and interpretability.

MobileNet: Efficiency and Real-Time Applicability

MobileNet, designed for lightweight applications, delivered faster inference with lower computational cost. Although its accuracy was slightly lower than ResNet-50 and VGG16, it proved highly effective for real-time and mobile-based deployment scenarios.

Its depthwise separable convolutions significantly reduce model size and computation, making it suitable for use in portable healthcare devices and smartphone-based diagnostic systems.

Quantitative Results and Interpretation

The experimental results of this study provide a comprehensive comparison of deep learning architectures in the context of non-invasive anemia detection. Among the evaluated models, **ResNet-50** emerged as the most superior architecture, achieving the highest accuracy and recall rates. This high level of performance is primarily due to its deep residual framework, which allows the model to learn intricate features of the conjunctiva that are critical for identifying hemoglobin deficiency. For diagnostic tools intended for clinical use, where

reliability and precision are the top priorities to ensure patient safety, ResNet-50 stands out as the most dependable choice.

On the other hand, **VGG-16** delivered a consistent and balanced performance across all evaluation metrics, including precision and F1-score. Although it follows a more traditional sequential architecture, its ability to maintain stable results across different sets of images makes it a strong baseline for medical image classification. However, while it is robust, its higher computational demand and larger memory footprint were noted during the testing phase, which may pose challenges for deployment on low-spec hardware compared to more modern, optimized networks.

Further analysis revealed that **MobileNet** is exceptionally well-suited for specific real-world constraints. It demonstrated excellent speed and computational efficiency while maintaining an acceptable level of accuracy for screening purposes.

The use of depthwise separable convolutions significantly reduces the model's complexity, allowing it to provide near-instant results. This efficiency highlights the importance of selecting a model based on the specific requirements of the application environment. For instance, while a clinical laboratory might prioritize the high-precision output of ResNet-50, a field-level screening application in a rural area would benefit more from the portability and speed offered by MobileNet.

Analysis of Training Behavior

The training and validation curves generated during the experimentation provide a comprehensive understanding of how each model—ResNet-50, VGG-16, and MobileNet—learned from the conjunctiva dataset. These curves are essential for diagnosing the learning behavior, as they illustrate the trade-off between the training performance and the model's ability to generalize on unseen validation data. By analyzing the trajectory of the accuracy and loss over 50 epochs, we can determine the stability and efficiency of the proposed deep learning architectures.

ResNet-50 demonstrated the most superior learning curve, characterized by a steady increase in accuracy and a consistent decline in loss. The convergence was remarkably smooth, with minimal fluctuations, which can be attributed to its residual learning framework. The skip connections within ResNet-50 allow the gradients to flow more effectively through the 50 layers, preventing the common issue of gradient saturation. Consequently, ResNet-50 achieved the lowest validation loss among all models, proving its high reliability for medical diagnostic tasks where precision is paramount.

In contrast, the **VGG-16** model, while providing a stable baseline, showed a slight gap between the training and validation curves. This behavior suggests a minor case of overfitting, likely due to the model's large number of trainable parameters and its relatively simpler sequential architecture compared to modern networks. While VGG-16 eventually reached a respectable accuracy level, its loss curve exhibited more noise and required more computational time to stabilize. This indicates that while VGG-16 is consistent, it lacks the advanced regularization benefits found in deeper architectures like ResNet.

Lastly, **MobileNet** displayed a unique learning pattern characterized by rapid initial convergence. Due to its depthwise separable convolutions, the model has fewer parameters to optimize, allowing it to reach an optimal accuracy level much faster than ResNet-50 or VGG-16. Although its final validation accuracy was slightly lower than the ResNet-50 benchmark, its curves remained closely aligned, indicating strong generalization. This makes MobileNet a highly efficient candidate for deployment in real-time scenarios where quick inference and low memory usage are required without significantly compromising the diagnostic quality.

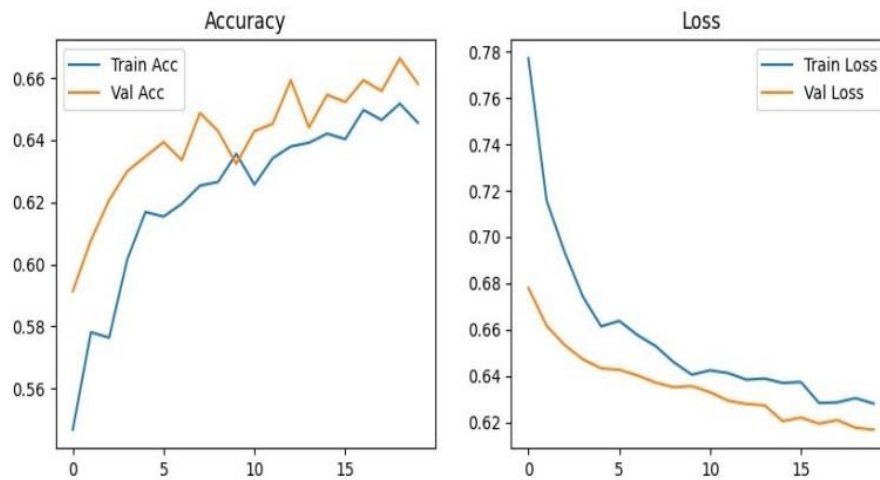


Fig. 4.1: Training Accuracy and Loss Graph

Visualization and Classification Insights

Visualization techniques such as confusion matrices reveal the classification performance of each model in detail. ResNet-50 showed the lowest number of false negatives, which is particularly important in medical diagnosis where missing a positive case can have serious consequences.

The models were also able to identify subtle differences in conjunctiva coloration, which is a key indicator of hemoglobin levels. This demonstrates the capability of deep learning models to detect patterns that may not be easily visible to the human eye.

		Predicted Class	
		Anemic	Non-Anemic
Actual Class	Anemic	96 TP	27 FP
	Non-Anemic	15 FN	144 TN

Fig. 4.2: Confusion Matrix of ResNet-50

Practical Implications in Healthcare

The results of this study have significant implications for modern healthcare systems. The proposed non-invasive approach eliminates the need for blood sample collection, reducing patient discomfort and risk of infection.

In rural and underserved areas where laboratory facilities are limited, this system can serve as an effective screening tool. By integrating the model into mobile applications, healthcare workers can perform quick and reliable anemia detection using just a smartphone camera. Furthermore, the system can support early diagnosis, enabling timely treatment and prevention of severe complications associated with anemia.

User Accessibility and Deployment Potential

One of the key strengths of the proposed system is its adaptability for real-world deployment. MobileNet, in particular, enables integration into mobile platforms due to its lightweight nature.

		MobileNet	
		Negative	Positive
Actual Class	Negative	74	26
	Positive	10	110
		Predicted Class	

Fig. 4.3: Confusion Matrix of MobileNet

The system can be deployed as:

- A mobile application for on-field diagnosis
- A web-based platform for remote healthcare services
- A clinical decision support tool for doctors

This flexibility ensures that the technology can reach a wide range of users, from healthcare professionals to patients in remote locations.

Future Enhancements

Future work can focus on enhancing the proposed anemia detection system by incorporating several advanced improvements and extensions. One important direction is the expansion of the dataset to include a wider range of demographic variations such as age, gender, ethnicity, and different lighting conditions, which will improve the model's generalization capability and robustness.

The system can also be extended by incorporating additional visual indicators such as nail bed, tongue, and skin images. This multi-modal approach can significantly improve diagnostic accuracy and enable the detection of other related medical conditions. Furthermore, the development of hybrid or ensemble models by combining multiple deep learning architectures can enhance prediction performance and reliability.

Improving image preprocessing techniques, including advanced normalization, noise reduction, and illumination correction, can further increase the accuracy of the system. In addition, integrating cloud-based platforms will enable large-scale deployment, remote access, and real-time processing, making the system more scalable and accessible.

With the rapid advancements in artificial intelligence and edge computing, the proposed system can be transformed into a real-time, portable, and intelligent healthcare solution. These developments will not only improve performance but also enhance accessibility, enabling widespread adoption in both urban and rural healthcare environments.

Conclusion of Results and Discussion

In conclusion, the experimental analysis clearly demonstrates that deep learning-based approaches are highly effective for non-invasive anemia detection using conjunctiva images. The comparative evaluation of different convolutional neural network models highlights that each architecture offers unique advantages in terms of accuracy, efficiency, and deployment feasibility.

Among the evaluated models, **ResNet-50** achieved the highest accuracy and overall performance due to its deep residual learning capability, making it highly suitable for clinical diagnostic applications where precision is critical. **VGG-16** provided stable and consistent results, while **MobileNet** proved to be highly efficient with faster processing speed and lower

computational requirements, making it ideal for real-time and mobile-based healthcare solutions.

The proposed system represents a significant advancement in the field of healthcare diagnostics by offering a non-invasive, cost-effective, and accessible alternative to traditional blood test-based methods. By leveraging deep learning and image analysis, the system enables early detection of anemia, reduces dependency on laboratory infrastructure, and enhances healthcare accessibility, especially in rural and resource-limited environments.

Overall, this work highlights the transformative potential of artificial intelligence in modern healthcare and paves the way for the development of intelligent, scalable, and patient-friendly diagnostic systems that can contribute to improved health outcomes and a more inclusive healthcare ecosystem.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

The rapid advancement of Artificial Intelligence (AI) and Deep Learning (DL) has significantly transformed modern healthcare systems by enabling automated, accurate, and efficient disease diagnosis. In this context, the proposed project presents a novel approach for **non-invasive anemia detection using conjunctiva images** and deep learning models, aiming to overcome the limitations of traditional diagnostic techniques.

Conventionally, anemia diagnosis relies on laboratory-based blood tests such as Complete Blood Count (CBC), which require blood sample collection, laboratory infrastructure, and trained medical professionals. Although these methods are highly accurate, they are often time-consuming, costly, and inaccessible in rural and resource-constrained environments. These challenges highlight the urgent need for alternative diagnostic solutions that are fast, affordable, and easily accessible.

The proposed system addresses these challenges by utilizing **image-based deep learning techniques** to detect anemia through analysis of the conjunctiva region of the eye. The conjunctiva serves as a reliable indicator of hemoglobin levels, as variations in its color and texture reflect the presence or absence of anemia. By leveraging this physiological characteristic, the system provides a completely non-invasive and user-friendly diagnostic approach.

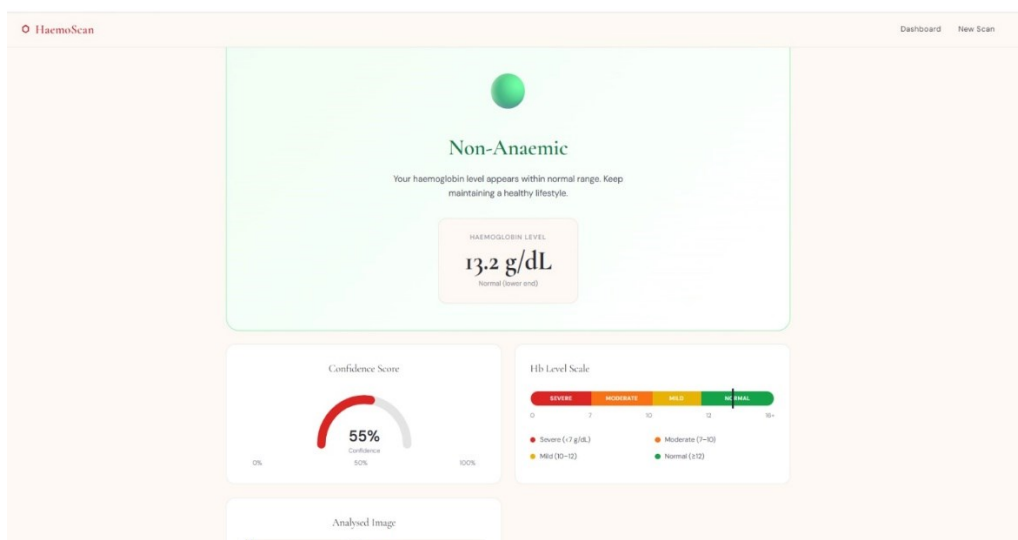


Fig. 5.1 : Result

The implementation of three advanced Convolutional Neural Network (CNN) architectures—ResNet-50, VGG-16, and MobileNet—enables robust feature extraction and accurate classification of conjunctiva images. Each model contributes uniquely to the system:

- **ResNet-50**, with its deep residual learning architecture, effectively captures complex patterns and achieves the highest classification accuracy.
- **VGG-16**, known for its simplicity and consistency, provides stable and reliable performance across different datasets.
- **MobileNet**, designed for efficiency, offers fast and lightweight computation, making it highly suitable for real-time and mobile-based applications.

The comparative analysis of these models demonstrates that deep learning techniques are highly effective for anemia detection. While ResNet-50 excels in accuracy, MobileNet provides a practical balance between performance and computational efficiency, making it ideal for deployment in real-world scenarios.

Furthermore, the system integrates essential components such as image preprocessing, feature extraction, and classification into a unified pipeline, ensuring smooth and efficient operation. The ability to deliver rapid predictions with minimal user input significantly reduces human effort and diagnostic delay.

The results obtained from the experimental analysis confirm that the proposed system achieves high accuracy and reliability in anemia detection. The system also demonstrates strong potential for scalability and adaptability, allowing it to be implemented in various healthcare settings, including rural clinics, mobile health units, and telemedicine platforms.

Overall, this project successfully establishes that **deep learning-based image analysis can serve as a powerful tool for non-invasive medical diagnosis**. It not only enhances diagnostic efficiency but also contributes to improving healthcare accessibility, particularly in underserved regions.

5.2 Future Scope

Although the proposed system demonstrates promising performance and practical applicability, there are several avenues for further enhancement and expansion to make it more robust, scalable, and clinically viable.

Integration with Mobile and Real-Time Systems

One of the most significant future developments involves transforming the system into a **mobile application** that allows users to capture conjunctiva images using smartphone cameras. Real-time processing capabilities can further enable instant diagnosis, making the system highly accessible and user-friendly.



Fig.5.1 : Future of Anemia Detection System

Expansion of Dataset and Model Generalization

The accuracy and robustness of deep learning models can be further improved by training on larger and more diverse datasets. Including images from different age groups, ethnicities, and lighting conditions will enhance the model's generalization capability and reduce bias.

Multi-Modal Disease Detection

The system can be extended to support **multi-disease detection** by incorporating additional image sources such as nail beds, tongue, and skin. This multi-modal approach can improve diagnostic accuracy and enable detection of other medical conditions such as jaundice and cyanosis.

Cloud-Based Healthcare Integration

Future work can focus on deploying the system on **cloud platforms**, enabling remote access, centralized data storage, and large-scale screening. This would allow healthcare professionals to monitor patients and access diagnostic results from anywhere.

Model Optimization and Ensemble Learning

Further improvements can be achieved by applying advanced techniques such as hyperparameter tuning, transfer learning optimization, and ensemble modeling. Combining multiple models can enhance prediction accuracy and robustness.

Integration with Clinical Decision Support Systems

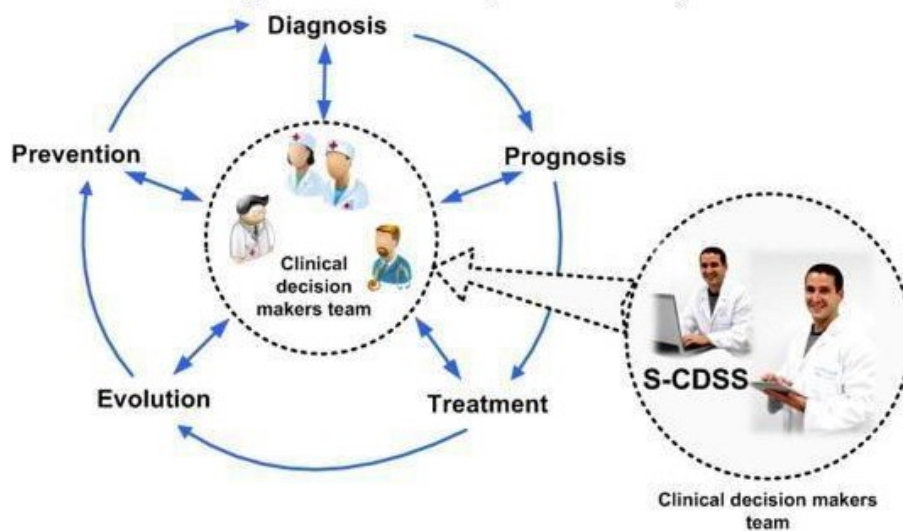


Fig. 5.2 : How CDSS Works

The proposed system can be integrated with hospital management systems and electronic health records (EHR) to support clinical decision-making. This integration can enhance the efficiency of healthcare workflows and improve patient management.

Explainable Artificial Intelligence (XAI)

Incorporating explainable AI techniques can make the system more transparent and trustworthy. Visualization tools such as heatmaps can help medical professionals understand how the model makes predictions, increasing confidence in its use.

Deployment in Rural Healthcare Systems

The system has strong potential for deployment in rural and underserved areas where access to laboratory facilities is limited. With proper optimization, it can be used as a screening tool for large populations.

5.3 Final Discussion

The development of the proposed anemia detection system represents a significant milestone in the shift toward intelligent, non-invasive, and accessible healthcare solutions. By integrating state-of-the-art deep learning architectures with medical image analysis, this system effectively bridges the gap between complex computational technology and practical clinical applications.

The ability to detect anemia through a simple image of the conjunctiva demonstrates how biomarkers can be digitized to provide a pain-free, needle-less diagnostic experience for patients worldwide.

The comparative performance of ResNet-50, VGG-16, and MobileNet highlights the versatility of the system. While the high precision of ResNet-50 ensures clinical-grade reliability, the lightweight nature of MobileNet opens doors for "Edge AI" deployment in resource-constrained environments. This flexibility ensures that the system is not limited to high-end

hospital servers but can function effectively on standard smartphones. Such a dual-approach is essential for addressing the global burden of anemia, particularly in regions where laboratory infrastructure is either unavailable or unaffordable.

As artificial intelligence continues to evolve, systems like this will play a crucial role in the era of preventive healthcare and early disease detection. Beyond just providing a binary diagnosis, these AI-driven tools can assist healthcare workers in monitoring patient outcomes over time through longitudinal data analysis.

The integration of AI into the medical field does not replace the expertise of a doctor; instead, it enhances their diagnostic capabilities and ensures that high-quality medical screening is accessible to a wider, more diverse population.

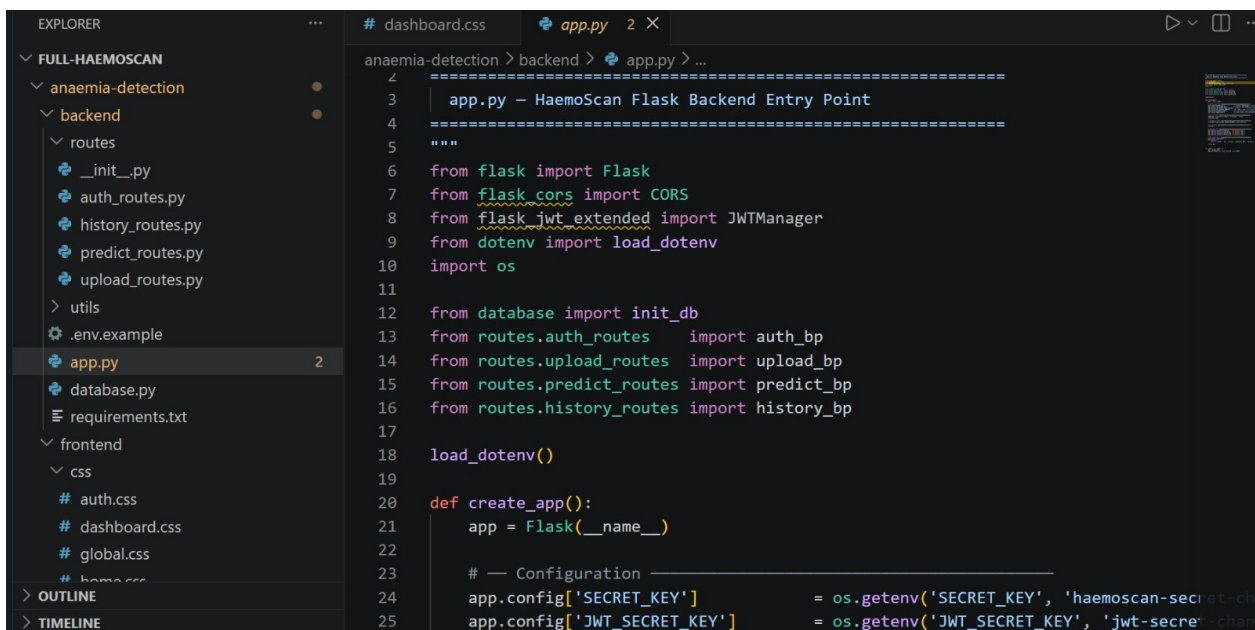
In conclusion, this project demonstrates that AI-driven healthcare solutions have the immense potential to revolutionize traditional diagnostic methods. By providing a faster, smarter, and more inclusive framework for screening, the system paves the way for a future where technology and medicine work in perfect synergy. The success of this model serves as a foundation for further research into multi-disease screening through ocular imaging, ultimately contributing to a more proactive and efficient global healthcare system.

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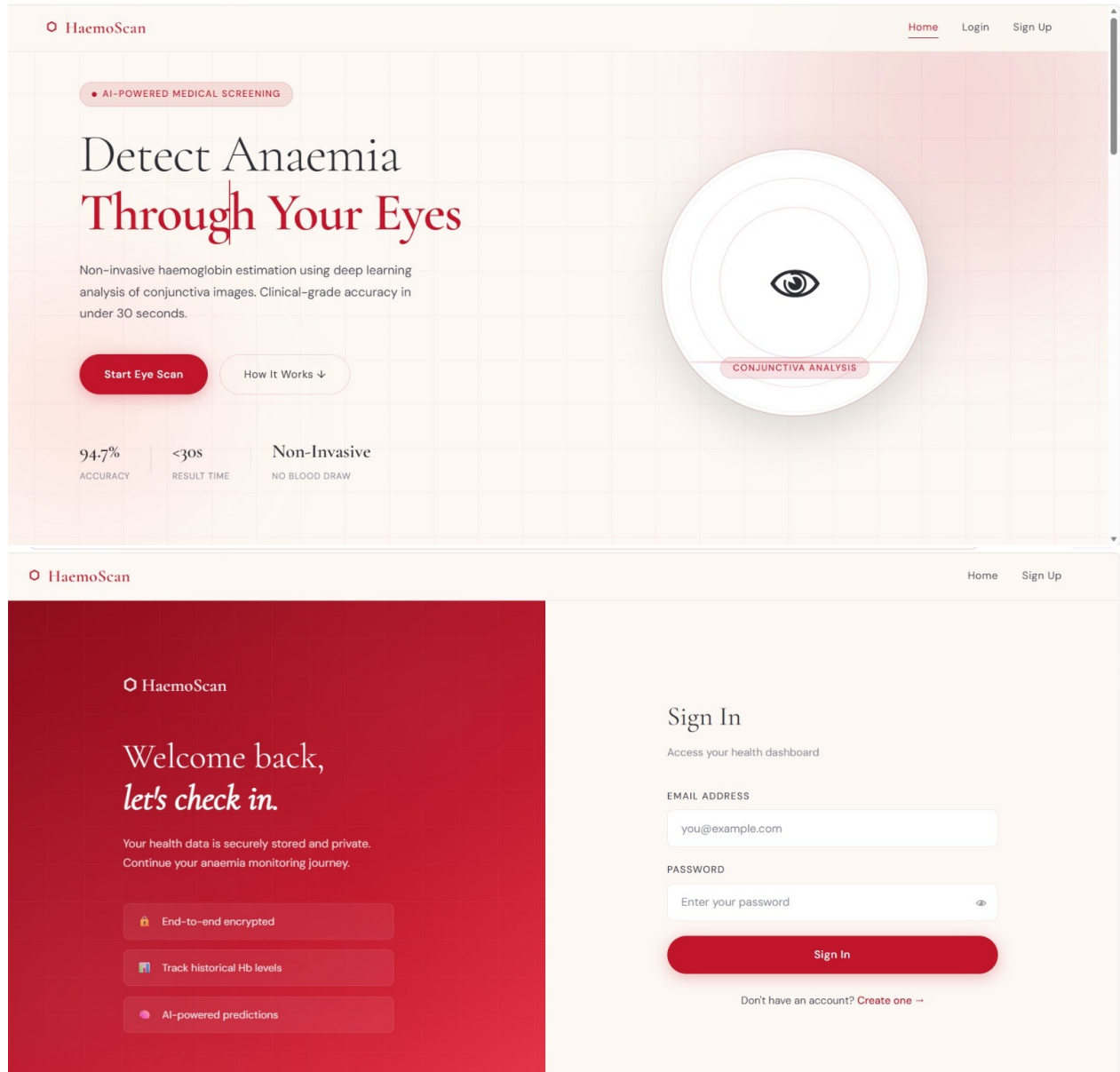
APPENDIX 1



The screenshot shows a code editor with a dark theme. On the left is the 'EXPLORER' sidebar showing a project structure for 'FULL-HAEMOSCAN'. The 'backend' folder is expanded, showing files like 'routes', 'utils', and 'app.py'. The 'app.py' file is selected and highlighted. The main editor area shows the content of 'app.py', which is a Flask application. The code includes imports for Flask, Flask-CORS, Flask-JWT-Extended, and dotenv. It also shows the initialization of the app, loading of routes, and configuration of environment variables.

```
# dashboard.css  app.py 2 X
anaemia-detection > backend > app.py > ...
4 =====
5 app.py - HaemoScan Flask Backend Entry Point
6 =====
7 """
8 from flask import Flask
9 from flask_cors import CORS
10 from flask_jwt_extended import JWTManager
11 from dotenv import load_dotenv
12 import os
13
14 from database import init_db
15 from routes.auth_routes import auth_bp
16 from routes.upload_routes import upload_bp
17 from routes.predict_routes import predict_bp
18 from routes.history_routes import history_bp
19
20 load_dotenv()
21
22 def create_app():
23     app = Flask(__name__)
24
25     # --- Configuration ---
26     app.config['SECRET_KEY'] = os.getenv('SECRET_KEY', 'haemoscan-secret-key')
27     app.config['JWT_SECRET_KEY'] = os.getenv('JWT_SECRET_KEY', 'jwt-secret-key')
```

DASHBOARD DESIGN OF WEB BASED APPLICATION OF ANEMIA DETECTION SYSTEM



PLAGIARISM REPORT



Page 2 of 70 - AI Writing Overview

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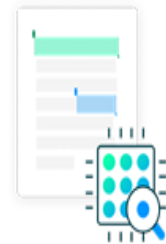
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Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

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To: Nandan Mishra <nandan2503mishra@gmail.com>

4 March 2026 at 13:56

Dear Author(s),

We are pleased to inform you that your paper entitled "An Early Detection of Anemia using MobileNet Architecture based Deep Neural Network", submitted to FIRST INTERNATIONAL CONFERENCE ON NEXUS OF DIGITALIZATION, INTELLIGENCE AND APPLICATIONS, has been accepted for presentation and publication in the conference proceedings, subject to the submission of the camera-ready version and final quality check by the internal screening committee.

Paper ID: 338

Paper Title: An Early Detection of Anemia using MobileNet Architecture based Deep Neural Network

Kindly log in to your Microsoft CMT account and upload the camera-ready paper. Please ensure that the revised manuscript strictly follows the conference formatting guidelines and incorporates the reviewers' comments, wherever applicable. Please go through the reviewers comments, modify the paper and upload the paper. You are requested to upload camera ready paper on or before 15 March 2026.

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